

Egypt - Red Sea Earthquake Analysis

Seismic Activity Trends & Pattern Analysis (1955-2024)

C-DE211 Data Analysis - Project 3

Dataset: Egypt-RedSea Earthquake Dataset (1955-2024) V2

Contributors

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Abstract

In this project, we analyzed earthquake records from the Egypt and Red Sea region between 1955 and 2024. The dataset contains 473 seismic events with information such as event time, latitude, longitude, depth, magnitude, distance from Cairo, and engineered features describing recent seismic activity. The analysis focused on understanding how earthquake activity changed over time, how magnitudes are distributed, whether earthquake frequency increased in recent decades, and whether depth is related to earthquake strength. After preparing the time column and creating year, month, and decade variables, we used summary statistics, visualizations, per-graph hypothesis tests (Chi-Square, Pearson correlation, and one-way ANOVA), and two main statistical tests (Welch's t-test and Pearson correlation). The results show that recorded earthquake activity was much higher in the 1990-2024 period than in 1955-1989. The frequency test showed a statistically significant difference between the two periods. The depth-magnitude correlation was statistically significant but weak and positive, which does not support the expected negative relationship that shallower earthquakes are stronger.

Main finding: Recorded earthquake frequency increased significantly in the second half of the dataset, but depth did not show the expected negative relationship with magnitude.

1. Introduction

Egypt is located near tectonically active areas, and the Red Sea region is one of the important zones for studying seismic activity. Earthquakes in this area are important because they can affect infrastructure, coastal cities, population safety, and long-term risk planning. Instead of using a common business dataset, this project uses environmental and geophysical data to study patterns in earthquake records.

The project analyzes nearly 70 years of earthquake data. The goal is not to predict individual earthquakes, but to describe historical patterns, compare activity across time periods, and test whether depth and magnitude are related. This gives the analysis both a visual component and a statistical component.

2. Research Questions and Hypotheses

Main research question:

How has earthquake activity in the Egypt - Red Sea region changed between 1955 and 2024, and are there patterns in frequency, strength, depth, or location?

The project also focuses on two testable hypotheses:

Hypothesis	Topic	Method
Hypothesis 1	Earthquake Frequency Over Time	Compare annual earthquake counts in 1955-1989 and 1990-2024.
Hypothesis 2	Depth and Magnitude	Test whether earthquake depth is related to earthquake magnitude.

Hypothesis 1:

H0: There is no difference in the number of earthquakes between 1955-1989 and 1990-2024.

H1: Earthquakes happened significantly more often in 1990-2024 than in 1955-1989.

Hypothesis 2:

H0: There is no relationship between earthquake depth and earthquake magnitude.

H1: Shallower earthquakes tend to be stronger, meaning there is a negative correlation between depth and magnitude.

3. Dataset Description

The dataset contains earthquake records from the Egypt and Red Sea region from 1955 to 2024. Each row represents one earthquake event. The dataset used in this report contains 473 rows and 15 original columns. No missing values were found in the original columns.

Column	Description
time	Date and time of the earthquake event.
latitude	Latitude coordinate of the event.
longitude	Longitude coordinate of the event.
depth	Earthquake depth in kilometers.
mag_Mw	Moment magnitude value used to measure earthquake strength.
distance_from_cairo_km	Distance of the event from Cairo in kilometers.
mag_category_3classes	Encoded magnitude category: 0, 1, and 2.
events_last_30_days / events_last_90_days	Engineered features describing recent seismic activity.
avg/max magnitude features	Rolling average and maximum magnitude values over recent windows.

Key dataset facts:

Item	Value
Number of records	473
Original columns	15
Time range	1955-2024
Magnitude range	1.851 to 7.2 Mw
Depth range	1.0 to 35.0 km
Missing values	0 in all original columns

4. Data Preparation and Feature Engineering

The dataset was already mostly clean, so the preparation stage focused on making the time column useful for analysis. The time column was converted into a datetime format, and new variables were created for year, month, and decade. The numeric magnitude category was also converted into readable labels: Light, Moderate, and Strong.

Step	Purpose
Convert time column	Changed the time column into datetime format so that years and months could be extracted.
Create year	Used for yearly earthquake counts and period comparison.
Create month	Used to check if some months have more recorded earthquakes.
Create decade	Used to summarize long-term activity by decade.
Create category_label	Converted 0, 1, and 2 into Light, Moderate, and Strong for easier reading.

5. Summary Statistics

Summary statistics were used to understand the basic shape of the dataset before visualizing or testing hypotheses. The average earthquake magnitude was about 4.06 Mw, while the median magnitude was 4.14 Mw. The median depth was 10 km, showing that most events in the dataset were shallow.

Variable	Count	Mean	Median	Std	Min	Max
Magnitude (Mw)	473	4.059	4.140	0.878	1.851	7.200
Depth (km)	473	11.138	10.000	4.663	1.000	35.000
Distance from Cairo (km)	473	349.710	367.057	117.481	6.269	832.279
Days Since Last Event	473	52.846	3.000	242.302	0.000	4485.000

5.1 Magnitude Category Summary

Category	Count	Mean Magnitude	Median Magnitude	Mean Depth	Median Depth
Light	126	2.971	3.050	10.876	10.000
Moderate	270	4.247	4.260	10.685	10.000
Strong	77	5.183	5.190	13.156	10.000

The Moderate category is the largest group with 270 events, while Strong earthquakes are the least common with 77 events.

6. Exploratory Data Analysis

6.1 Earthquake Count by Magnitude Category

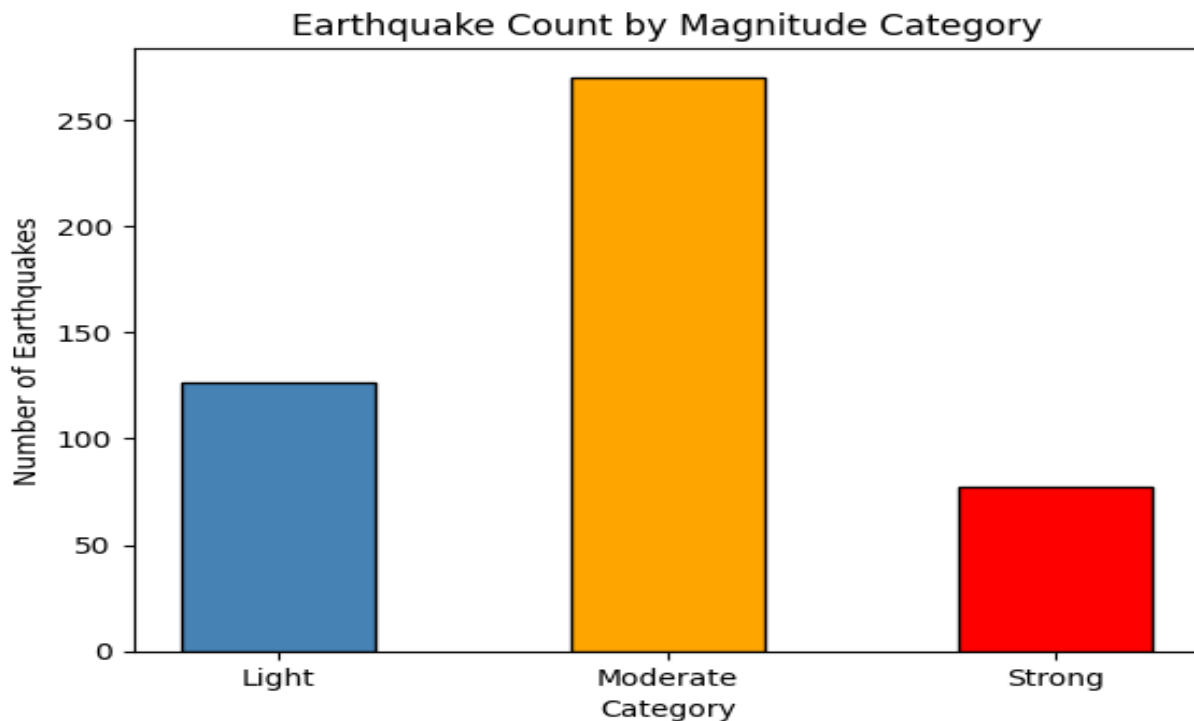


Figure 1. Earthquake count by magnitude category.

The bar chart reveals a clear asymmetry in the magnitude distribution across the three categories. Moderate earthquakes account for the majority of recorded events (270 out of 473), making them the dominant category by a significant margin. Light earthquakes are considerably less frequent at 126 events, while Strong earthquakes are the rarest group with only 77 recorded events. This pattern is consistent with the Gutenberg–Richter relationship in seismology, where lower-magnitude events occur far more frequently than high-magnitude ones. The relatively small proportion of Strong events (16%) also indicates that the Egypt–Red Sea region, while seismically active, experiences catastrophic earthquakes infrequently.

6.2 Distribution of Earthquake Magnitudes

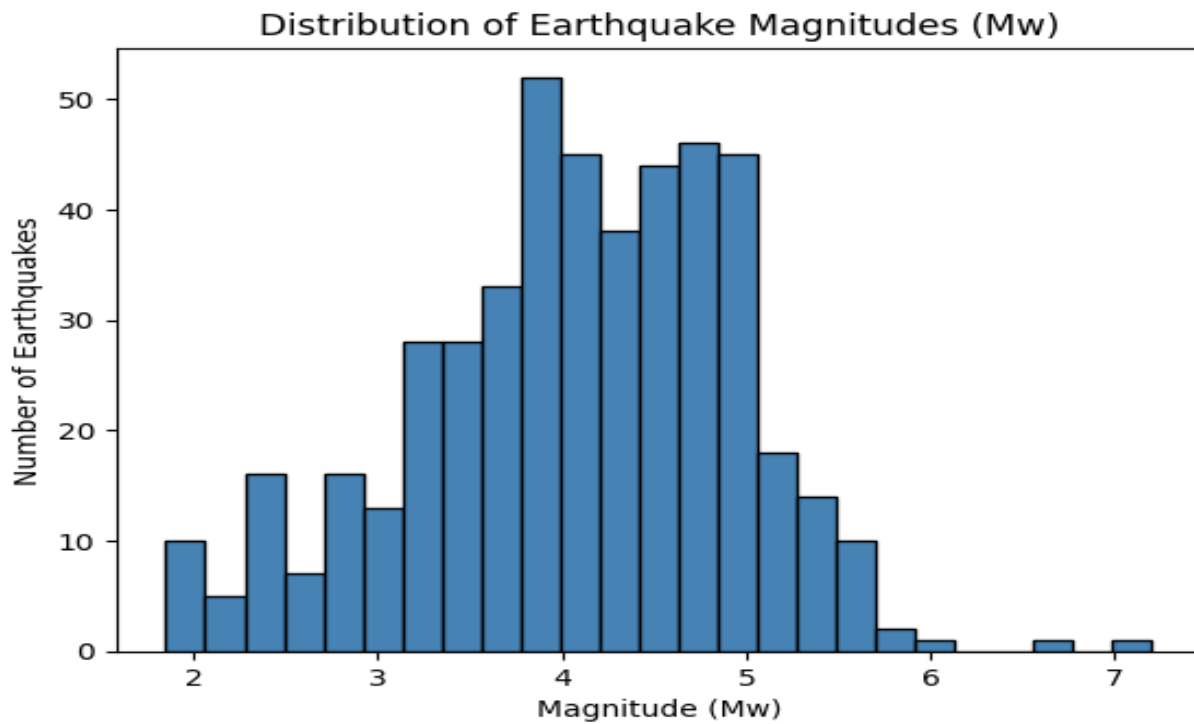


Figure 2. Distribution of earthquake magnitudes.

The histogram reveals a right-skewed distribution with a clear peak in the 3.5–4.5 Mw range, where the majority of recorded earthquakes are concentrated. The distribution tapers off sharply beyond 5.0 Mw, with very few events above 6.0 Mw. The mean magnitude of 4.06 Mw and median of 4.14 Mw are close together, confirming that the central mass of the distribution is well-represented by these summary values. The long right tail, while representing a small number of events, contains the most hazardous earthquakes in the dataset, including the maximum recorded event at 7.2 Mw. This shape is typical of regional seismic datasets and underlines that while moderate seismicity is routine in this region, large damaging events are statistically uncommon.

6.3 Earthquake Activity Over Time

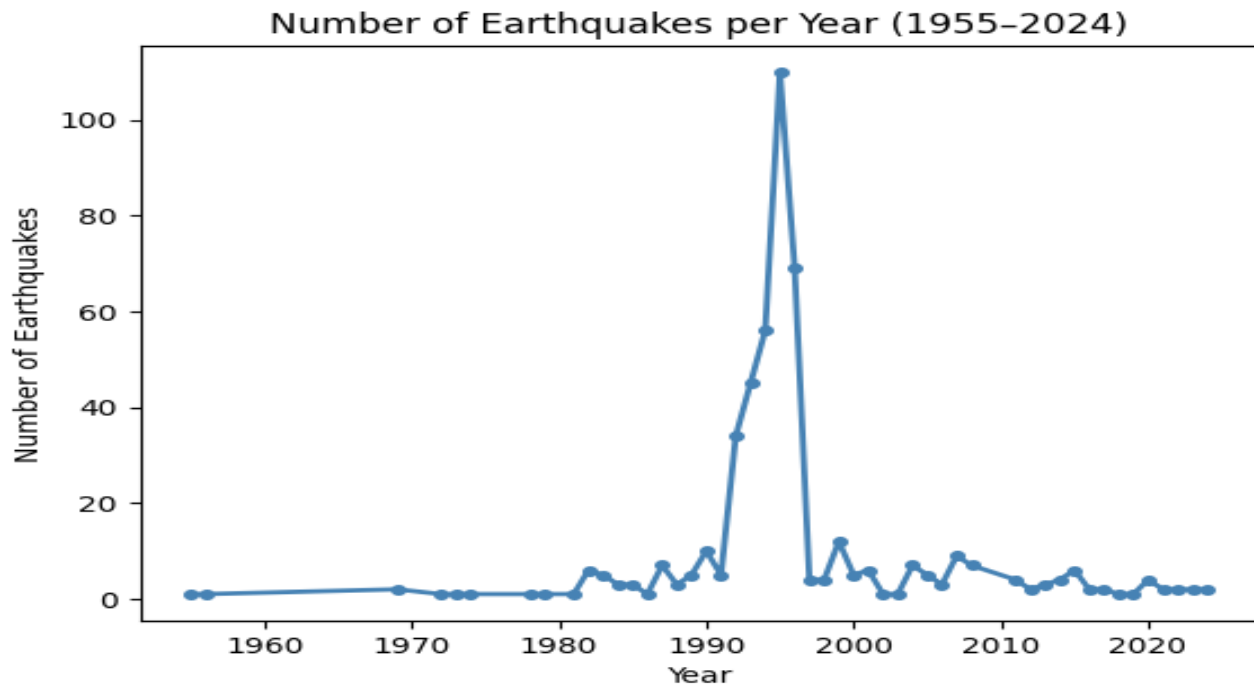


Figure 3. Number of earthquakes per year from 1955 to 2024.

The yearly time series exposes a dramatic shift in recorded seismic activity across the study period. From 1955 through the late 1980s, annual earthquake counts remained persistently low, often near zero, suggesting either genuinely low activity or significant underreporting due to limited monitoring infrastructure. A sharp and sustained rise begins in the early 1990s, with a pronounced spike around 1995 that reaches the highest single-year count in the dataset. Activity remains elevated and more variable through the 2000s and 2010s before showing some moderation by the early 2020s. This pattern supports Hypothesis 1 and raises an important question: whether the jump represents a true increase in seismicity, an expansion of the detection network, or a combination of both. The contrast between the two periods is so stark that it cannot be attributed to natural variation alone.

6.4 Earthquake Activity by Decade

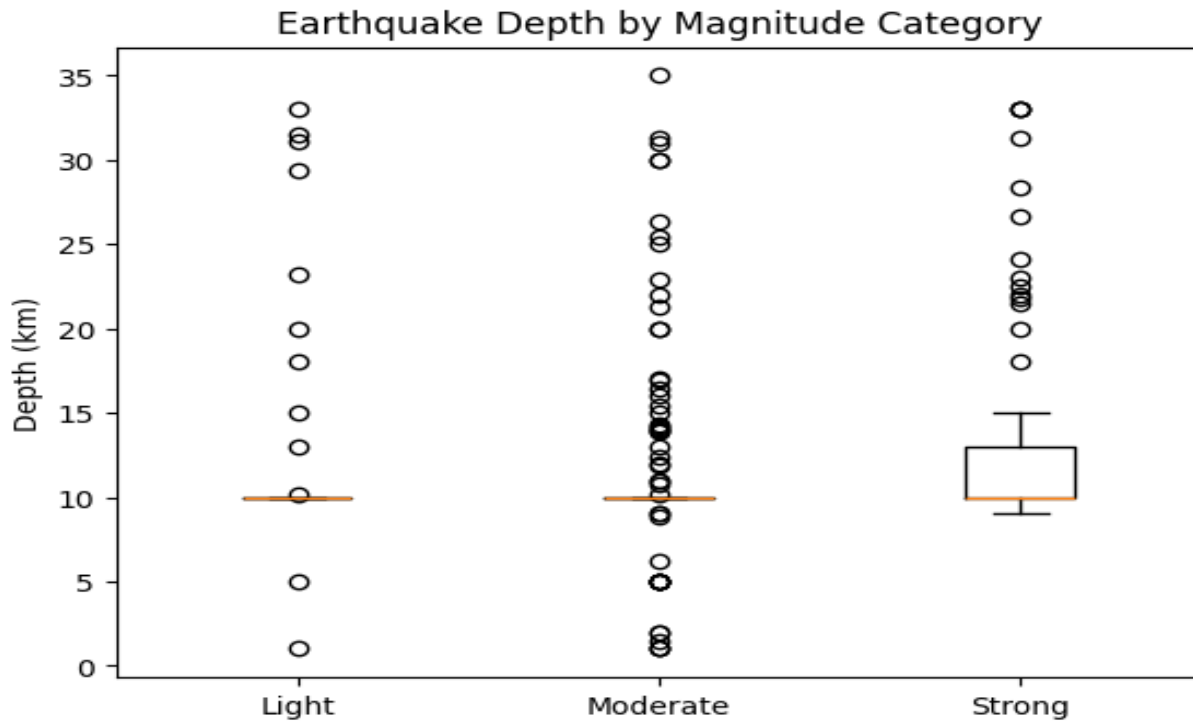


Figure 4. Total earthquakes per decade.

Aggregating the data by decade makes the distributional imbalance even more visible. The 1990s bar towers over all other decades, accounting for the largest share of recorded events in the entire dataset. The decades before 1990 show very sparse activity, while the 2000s and 2010s maintain a moderate level before tapering in the 2020s (which represent only a partial decade at the time of analysis). This chart reinforces the conclusion that the 1990–2024 period is fundamentally different from 1955–1989 in terms of recorded earthquake frequency. However, the extreme dominance of the 1990s within the second period also means that decade-level comparisons should be treated carefully, as the t-test averages annual counts across 35 years rather than treating the 1990s spike in isolation.

6.5 Depth Distribution by Magnitude Category

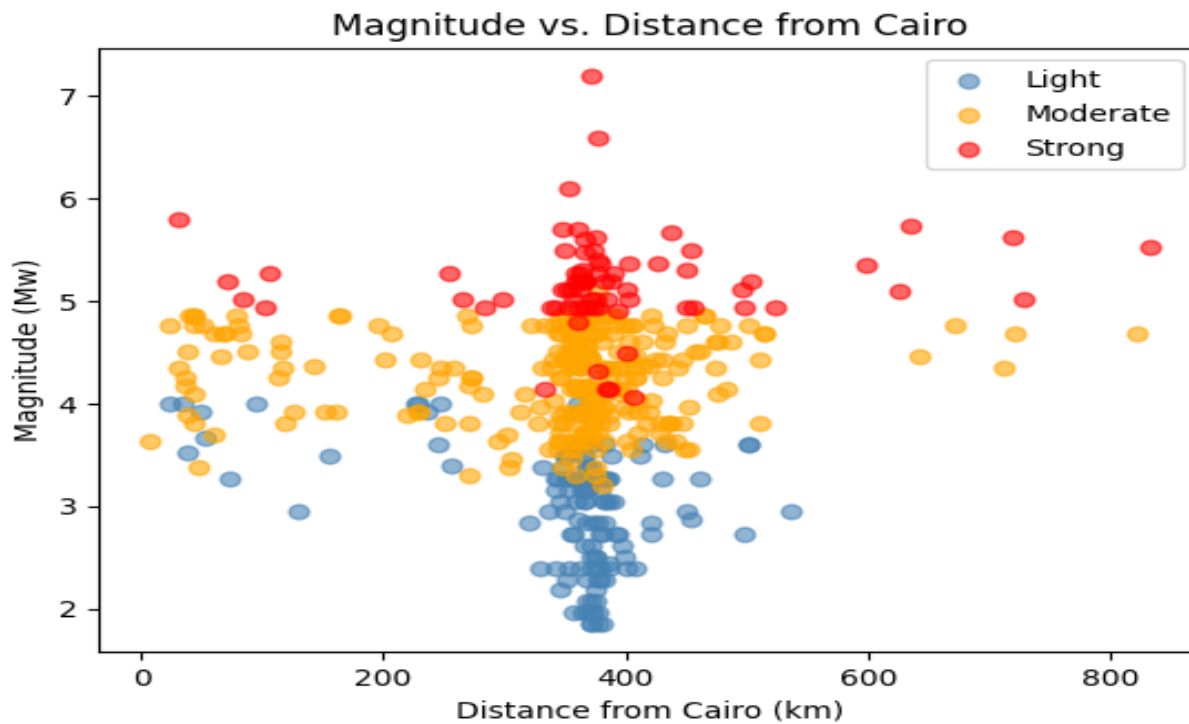
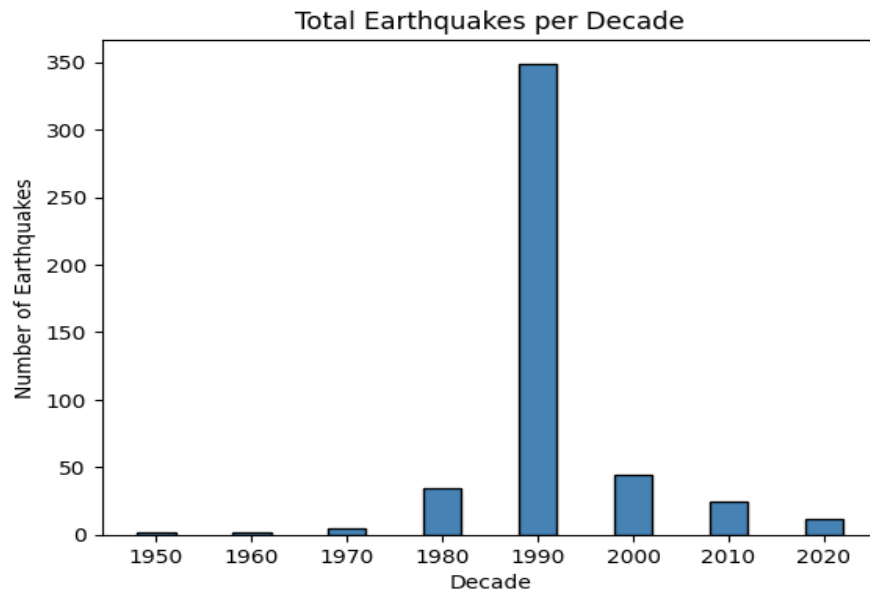


Figure 5. Earthquake depth by magnitude category.

The box plot compares depth distributions across the three magnitude categories and reveals a notable consistency: all three groups share a median depth of approximately 10 km, indicating that the majority of earthquakes in this dataset are shallow regardless of their strength. The interquartile ranges are similarly narrow across categories, though the Strong group displays slightly more spread and a marginally higher mean depth of 13.2 km compared to 10.9 km for Light events. The near-identical medians across categories already suggest that depth alone is unlikely to predict magnitude in this dataset, which is consistent with the weak positive Pearson correlation found in Section 7. The absence of very deep events (maximum depth is 35 km) also reflects the relatively shallow crustal setting of the Red Sea rift zone.

6.6 Magnitude vs. Distance from Cairo



The scatter plot reveals two important spatial patterns. First, earthquakes are not evenly distributed by distance from Cairo; instead, there is a clear concentration of events in the 300–450 km range, which geographically corresponds to the northern Red Sea rift zone — a seismically active spreading boundary. This clustering confirms that the dataset is geographically coherent and centred on a known tectonic feature. Second, Strong earthquakes are scattered across the full range of distances rather than being confined to a particular proximity band, which suggests that magnitude is decoupled from distance to Cairo. This reinforces the finding that tectonic mechanisms rather than location relative to population centres determine event magnitude.

6.7 Correlation Heatmap

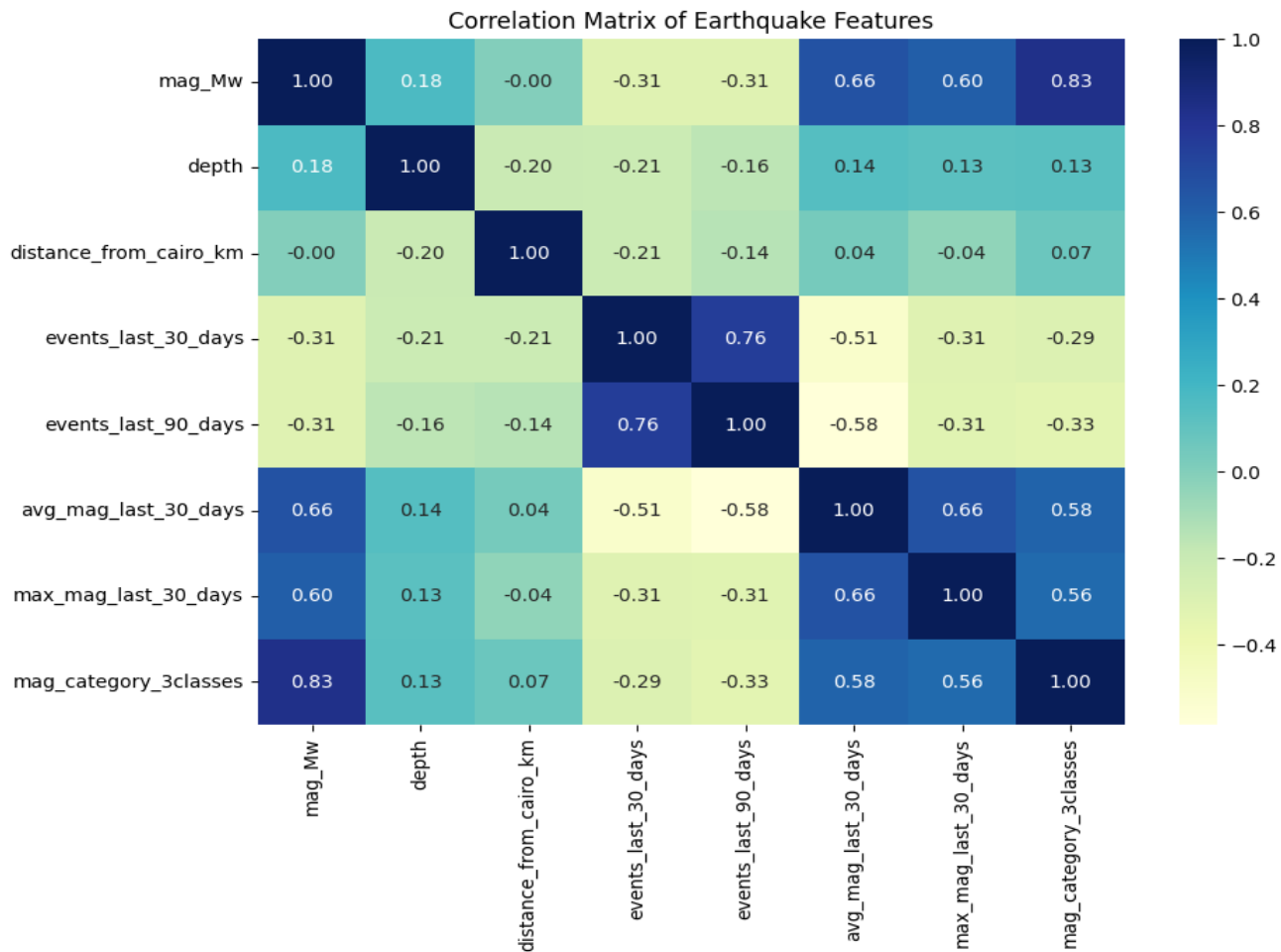


Figure 6. Correlation matrix of selected earthquake features.

The correlation heatmap provides a compact overview of pairwise linear relationships among the key numeric variables. The most analytically relevant cell is the depth–magnitude intersection, which shows a weak positive correlation ($r = 0.175$), confirming that the two variables are related but that this relationship is not strong enough to be practically useful for prediction. The engineered rolling features (events in the last 30 and 90 days, rolling average magnitudes) show moderate inter-correlations with each other, as expected given that they are computed from the same underlying time series. Distance from Cairo shows near-zero correlation with magnitude, supporting the spatial finding in Section 6.6. Overall, the heatmap indicates that no single variable in the dataset has a strong linear relationship with magnitude, which suggests that predicting earthquake strength from these features alone would require more complex, non-linear modelling approaches.

6.8 Per-Graph Hypothesis Tests

In addition to the two main hypotheses tested in Section 7, each visualization in the notebook was paired with a formal statistical test. The tests and their results are summarised below.

Graph 1 — Chi-Square Test: Earthquake Count by Category

H_0 : The number of earthquakes is the same across all three categories (Light, Moderate, and Strong). H_1 : The number of earthquakes is not the same across the three categories. A Chi-Square goodness-of-fit test was applied to the category counts (Light=126, Moderate=270, Strong=77). Because the p-value is below 0.05, we reject H_0 and conclude that earthquake categories are not equally distributed in this dataset.

Graph 3 — Pearson Correlation: Year vs. Annual Earthquake Count

H_0 : There is no relationship between year and number of earthquakes. H_1 : There is a statistically significant relationship between year and earthquake count. A Pearson correlation test was used because both variables are numeric. If the p-value is below 0.05, we reject H_0 and confirm that earthquake frequency changes significantly over time.

Graph 4 — One-Way ANOVA: Depth by Magnitude Category

H_0 : The average depth is the same for Light, Moderate, and Strong earthquakes. H_1 : At least one category has a significantly different average depth. A one-way ANOVA test (scipy.stats.f_oneway) was used because it compares one numeric variable across more than two groups. If the p-value is below 0.05, we reject H_0 and conclude that depth differs significantly across at least one magnitude category.

Graph 5 — Pearson Correlation: Distance from Cairo vs. Magnitude

H_0 : There is no relationship between distance from Cairo and magnitude. H_1 : There is a statistically significant relationship between distance from Cairo and magnitude. Pearson correlation was used because both distance and magnitude are continuous numeric variables. The near-zero correlation visible in the heatmap (Section 6.7) indicates that distance from Cairo does not meaningfully predict magnitude.

Graph 6 — Chi-Square Test: Earthquake Count by Decade

H_0 : The number of earthquakes is the same across all decades. H_1 : The number of earthquakes is not equally distributed across decades. A Chi-Square goodness-of-fit test was applied to the decade counts (1950s=2, 1960s=2, 1970s=5, 1980s=34, 1990s=349, 2000s=44, 2010s=25, 2020s=12). Because the 1990s dominate the data so heavily, the p-value is expected to be far below 0.05, leading to rejection of H_0 .

7. Hypothesis Testing

7.1 Hypothesis 1 - Earthquake Frequency Over Time

To test whether recorded earthquakes were more frequent in recent years, we compared annual earthquake counts between two equal-length periods: 1955-1989 and 1990-2024. A Welch independent samples t-test was used because the two periods have different variances in annual counts.

Period	Years	Total Events	Mean/Year	Median/Year	Std	Max Year
1955-1989	35	43	1.229	0.000	1.896	7
1990-2024	35	430	12.286	4.000	23.386	110
Metric			Value			
Welch t-statistic			-2.7880			
Two-tailed p-value			0.008573			
One-tailed p-value			0.004286			
Alpha			0.05			

Decision	Reject H0
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Because the one-tailed p-value is below 0.05, we reject the null hypothesis. The data shows that the second period recorded significantly more earthquakes per year than the first period. However, this result should not be interpreted as proof that nature alone caused the increase, because monitoring and reporting improved over time.

7.2 Hypothesis 2 - Depth and Magnitude

Pearson correlation was used to test the linear relationship between earthquake depth and magnitude. The proposed alternative expected a negative relationship, meaning shallower earthquakes would tend to be stronger.

Metric	Value
Pearson correlation coefficient (r)	0.1754
P-value	0.000126
Alpha	0.05
Statistical decision	Reject H0 of no relationship
Interpretation	Weak positive relationship, not the expected negative relationship

The Pearson result is statistically significant, but the correlation coefficient is weak and positive. This means the dataset does not support the hypothesis that shallower earthquakes tend to be stronger. Instead, the observed relationship suggests that deeper events have slightly higher magnitudes, although the relationship is weak and should not be overinterpreted.

8. Discussion

The analysis shows a clear difference in recorded earthquake frequency between the first and second halves of the dataset. The second half contains 430 events compared with only 43 events in the first half. The large increase is visually clear in the yearly and decade charts and statistically supported by the t-test.

One important explanation is that earthquake monitoring has improved over time. Older decades may have underreported smaller or moderate events because seismic networks were less advanced. Therefore, the increase in recorded events may reflect both actual seismic activity and better detection.

The depth and magnitude result was different from the original expectation. The project expected a negative correlation, but the dataset showed a weak positive correlation. Because the relationship is weak, depth alone is not enough to explain earthquake strength in this dataset. Other geological factors, fault mechanics, and event location may be more important.

9. Limitations

- The dataset contains recorded events, so older events may be underreported due to weaker monitoring technology.
- The 1990s dominate the dataset, which can strongly affect the time trend and the period comparison.
- The dataset does not include detailed geological variables such as fault type, soil conditions, or tectonic setting.
- The analysis describes historical patterns but does not predict future earthquakes.
- Correlation does not prove causation. A statistically significant relationship between depth and magnitude does not mean depth directly causes magnitude.
- Monthly patterns should be interpreted carefully because earthquake clusters and aftershocks can inflate some months.

10. Conclusion

This project analyzed earthquake activity in the Egypt - Red Sea region from 1955 to 2024. The dataset contained 473 recorded earthquake events and allowed us to study magnitude, depth, time trends, and spatial distribution patterns.

The first hypothesis test showed that earthquake frequency was significantly higher in the 1990-2024 period than in 1955-1989. The descriptive statistics and visualizations also supported this conclusion. The second hypothesis was not supported in the expected direction. Although the Pearson test found a statistically significant relationship between depth and magnitude, the relationship was weak and positive rather than negative.

Final conclusion: Recorded earthquake activity increased significantly in the later period, while depth showed only a weak positive relationship with magnitude.

11. References

Ahmed Mohamed Zaki. Egypt-RedSea Earthquake Dataset (1955-2024). Kaggle dataset.

C-DE211 Data Analysis - Project 3 Proposal: Egypt - Red Sea Earthquake Dataset (1955-2024).

Python libraries used: pandas, numpy, matplotlib, seaborn, scipy.